Qianou Christina Ma

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Research Focus

My research lies at the intersection of Human-AI Interaction, Generative AI, and the Learning Sciences. I **design, build, and evaluate LLM applications** to foster AI literacy, optimize human-AI task delegation, and support effective human-AI workflows, especially in **programming** and **computing education**. I conduct **mixed- methods research** to investigate how people collaborate and learn with AI systems, and design systems and frameworks to support training and evaluation of effective human-AI teams.

Education

Ph.D. in Human-Computer Interaction *Sept. 2022 - now*

Carnegie Mellon University (CMU) *Advisors:* Dr. **Ken Koedinger** & Dr. **Sherry Tongshuang Wu** *Pittsburgh, PA*

Dissertation: Training for the Future: Preparing Humans for Effective Human–AI Programming

Thesis Committee: Dr. Sherry Tongshuang Wu, Dr. Ken Koedinger, Dr. Brad Myers, Dr. Elena Glassman

M.S. in Human-Computer Interaction *Dec. 2024*

Carnegie Mellon University (CMU) *Pittsburgh, PA*

Relevant Coursework: Deep Learning; Advanced NLP; LLM Systems; AI Tools for SDE; Learning Media Design; Persuasive Design

B.S. in Computer Science & B.S. in Cognitive Science *May 2022*

Carnegie Mellon University (CMU) Minor in Design for Learning. QPA: 3.91/4.0 *Pittsburgh, PA*

Research & Work Experience

Applied Science Intern, Microsoft *Redmond, WA*

Mentors: Dr. Shamsi Iqbal, Dr. Siddarth Suri & Dr. Pranav Khape *2026 Summer*

- Leading *an empirical study* (paper in submission [P1]) on how expertise shapes agentic work and translating research to AI upskilling insights.
- Analyzing *human profiles* and *interaction behaviors and trajectories* to identify factors that predict a better future of work with LLM agents.

PhD Researcher, Carnegie Mellon University *Pittsburgh, PA*

Advisors: Dr. Ken Koedinger & Dr. Sherry Tongshuang Wu *2022 - present*

- Designed and built new measures and LLM-based learning tools to develop AI literacy resilient to model updates (see summary at [W1]). E.g., a teachable agent for CS1 debugging (HypoCompass [P12]) and a tutor to help end-users program in prompts (ROPE [P10]).
- Led *mixed-methods research* (literature reviews [W7, P11], controlled experiments [P10], classroom study [P8], and pre-post user studies [P12, P6], and conducted statistical modeling, log analysis, and qualitative coding to evaluate human-AI collaboration and learning effectiveness.
- Collaborated with *interdisciplinary teams* to co-author 10+ peer-reviewed papers (e.g., NLP [P5, W6], CSEd [P14, W5], SWE [P9], HCI [P8]), including a first-authored Best Paper [P12] and Honorable Mention [P13] and multiple invited presentations (IJCAI, CSCL).

Selected Publications

Manuscripts and Pre-prints

[P1] **Qianou Ma**, Pranav Khape, Rock Pang, Ken Koedinger, Tongshuang Wu, Siddarth Suri, and Shamsi Iqbal. Impact of Expertise and Trust on AI-assisted Work. In *Submission*, 2027.

[P2] Yeon Su Park, Nadia Arvi, Hae Ri Lee, Sehoon Lim, **Qianou Ma**, and Juho Kim. The Work Behind Delegation: A Framework for Supervising AI Coding Agents. In *Submission*, 2027.

[P3] Zora Zhiruo Wang, Apurva Gandhi, Rulin Shao, Aspen Chen, Jonas Mueller, Zhiqi Liang, Jett Chen, Michael Ryan, **Qianou Ma**, Luxi He, et al. Efficient Test-Time Adaptation through Human-AI Interaction. In *NeurIPS TTCL Workshop (accepted)*, 2026.

[P4] Xuhui Zhou*, Weiwei Sun*, Weihua Du, Jiarui Liu, Haojia Sun, **Qianou Ma**, Tongshuang Wu, Yiming Yang, and Maarten Sap. OdysSim: Building Foundation Models for Human Behavior Simulation. In *NeurIPS (accepted)*, 2026. **Equal contribution*.

[P5] Xuhui Zhou*, Weiwei Sun*, **Qianou Ma**, Yiqing Xie, Jiarui Liu, Weihua Du, Sean Welleck, Yiming Yang, Graham Neubig, Tongshuang Wu, et al. Mind the Sim2Real Gap in User Simulation for Agentic Tasks. In *COLM (accepted)*, 2026. **Equal contribution*.

Peer-reviewed Conference and Journal Publications

[P6] **Qianou Ma**, Megan Chai, Yike Tan, Jihun Choi, Jini Kim, Erik Harpstead, Geoff Kauffman, and Tongshuang Wu. "GenAI Defaults to Bias!" Gamify AI literacy through Reflections on Prompts. In *International Conference on Artificial Intelligence in Education (AIED)*, 2026. Code: </>.

[P7] **Qianou Ma***, Keyu He*, Valerie Chen, Wayne Chi, and Tongshuang Wu. RECAP: An End-to-End Platform for Capturing, Replaying, and Analyzing AI-Assisted Programming Interactions. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, 2026. Code: </>. **Equal contribution*.

[P8] **Qianou Ma**, Ken Koedinger, and Tongshuang Wu. Not Everyone Wins with LLMs: Behavioral Patterns and Pedagogical Implications for AI Literacy in Programmatic Data Science. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, 2026. Code: </>.

[P9] Chenyang Yang, Yike Shi, **Qianou Ma**, Michael Xieyang Liu, Christian Kästner, and Tongshuang Wu. What Prompts Don't Say: Understanding and Managing Underspecification in LLM Prompts. In *Findings of the Association for Computational Linguistics: ACL*, 2026.

[P10] **Qianou Ma**, Weirui Peng, Chenyang Yang, Hua Shen, Kenneth Koedinger, and Tongshuang Wu. What Should We Engineer in Prompts? Training Humans in Requirement-Driven LLM Use. In *Transactions on Computer-Human Interaction (TOCHI)*, 2025. Presented at CHI'26. Media coverage: SCS News; New York Times. Code: </>.

[P11] **Qianou Ma***, Dora Zhao*, Xinran Zhao, Chenglei Si, Chenyang Yang, Ryan Louie, Ehud Reiter, Diyi Yang, and Tongshuang Wu. SPHERE: An Evaluation Card for Human-AI System Evaluation. In *ACL Findings*, 2025. Media coverage: MIT AHA Workshop. Code: </>. **Equal contribution*.

[P12] **Qianou Ma**, Hua Shen, Kenneth Koedinger, and Tongshuang Wu. How to Teach Programming in the AI Era? Using LLMs as a Teachable Agent for Debugging. In *International Conference on Artificial Intelligence in Education (AIED)*, 2024. 🏆 **Best Paper** 🏆 **Best Interactive Event**. Invited presentations at IJCAI'25 and CSCL'25. Media coverage: The Link, New York Times. Code: </>.

[P13] Atharva Naik, Jessica Ruhan Yin, Anusha Kamath, **Qianou Ma**, Sherry Tongshuang Wu, Charles Murray, Christopher Bogart, Majd Sakr, and Carolyn P. Rose. Generating Situated Reflection Triggers About Alternative Solution Paths: A Case Study in Generative AI for Computer-Supported Collaborative Learning. In *International Conference on Artificial Intelligence in Education*. Springer, 2024. 🏆 **Best Paper Nominee**.

[P14] Atharva Naik, Jessica Ruhan Yin, Anusha Kamath, **Qianou Ma**, Sherry Tongshuang Wu, R. Charles Murray, Christopher Bogart, Majd Sakr, and Carolyn P. Rose. Providing tailored reflection instructions in collaborative learning using large language models. *British Journal of Educational Technology*:20, 2024.

Peer-reviewed Workshop, Posters & Work-in-Progress

[W1] **Qianou Ma**. Training for the Future: Preparing Humans for Effective Human–AI Programming. In *CHI EA'26: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (Doctoral Consortium)*, 2026.

[W2] **Qianou Ma**, Sherry Tongshuang Wu, and Kenneth R. Koedinger. What Calculus Should We Teach in the Age of AI? Identifying and Improving Higher-Order Thinking in College Math. In *Artificial Intelligence in Education Doctoral Consortium.* 2026.

[W3] Ruiwei Xiao, Shan Zhang, Xinying Hou, Ying-Jui Tseng, **Qianou Ma**, Yash Tadimalla, Qing Xiao, Jionghao Lin, Bo Jiang, John Stamper, and Kenneth R. Koedinger. AI Literacy For All: 2nd International Workshop on AI Literacy Education For All. In *Artificial Intelligence in Education FoL Workshops.* 2026.

[W4] **Qianou Ma**, Anika Jain*, Jini Kim*, Megan Chai*, and Geoff Kauffman. ImaginAItion: Promoting Generative AI Literacy Through Game-Based Learning. In *CHI EA'25: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, 2025. **Equal contribution*.

[W5] Aysa Xuemo Fan, Rully Agus Hendrawan, Yang Shi, and **Qianou Ma**. Enhancing Code Tracing Question Generation with Refined Prompts in Large Language Models. *SIGCSE TS (55th ACM Technical Symposium on Computer Science Education)*, 2024.

[W6] **Qianou Ma***, Yann Hicke*, Anmol Agarwal*, and Paul Denny. AI-TA: Towards an Intelligent Question-Answer Teaching Assistant using Open-Source LLMs. In *Proceedings of the Workshop on GenAI for Education @NeurIPS*, 2023. **Equal contribution*.

[W7] **Qianou Ma**, Tongshuang Wu, and Kenneth Koedinger. Is AI the better programming partner? Human-Human Pair Programming vs. Human-AI pAIr Programming. In *Proceedings of the Workshop on LLMs@AIED*, 2023. Presented at ICER'23.

Awards & Fellowships

Honors & Awards

2026	**Fellow on AI & Society**, Block Center for Technology and Society	*Pittsburgh, PA*
2024	**Best Paper** , top-1 best paper [P12] awarded at AIED 2024	*Recife, Brazil*
2024	**Best Interactive Event**, top-1 best interactive event [P12] at AIED 2024	*Recife, Brazil*
2024	**Best Paper Nominee**, top-6 best paper nominee [P13] at AIED 2024	*Recife, Brazil*
2023	**Generative AI Innovation Incubator (GAI3) Hackathon**, 1st place team awarded with $20,000	*Pittsburgh, PA*
2021	**Phi Beta Kappa**, 1 of 16 seniors selected in Fall 2021	*Pittsburgh, PA*

Fellowships & Fundings

2026	***What Calculus Should We Teach in the Age of AI? Identifying and Improving Higher-Order Thinking in Math***, Learnvia Seed Grant (Role: Co-PI; led proposal writing)	*$150,000*
2025	***RealEduArena: Evaluating LLMs' Pedagogical Implementations in Real AI Tutoring***, Google Gift Fund on Evaluations and Benchmarks (Role: lead PhD researcher)	*$75,000*
2025	***ImaginAItion: Promoting Generative AI Literacy Through Game-Based Learning***, CMU GSA/Provost Conference Fund	*$1500*
2024	***Explicitly Train Humans towards Human-AI Collaboration***, Google Academic Research Award (Role: lead PhD researcher; contributed to proposal writing)	*$100,000*
2021	***Support Designer-Teacher Collaboration in Educational Game Design Using Learning Science Principles***, CMU Dietrich College Senior Honors Fellowship	*$6500*
2020	***The contribution of local and global language statistics to developing structured semantic networks***, CMU Summer Undergraduate Research Fellowship	*$3500*
2020	***Improving Web Visualization to Support Teachers & Parents: a Case Study of the Carnegie Museum of Art***, CMU Small Undergraduate Research Grant	*$500*
2019	***Linguistic patterns differentially predict children's semantic structure relatedness***, Ireland Undergraduate Research Fund	*$1750*

Presentations & Talks

2026 Summer	**Invited Talk**, The Impact of Expertise on AI-assisted Work	*Microsoft*
2026 Spring	**Invited Talk**, What to Teach and How to Teach Programming in the AI Era Penn; Cornell; Columbia; UMich; UChicago; UIUC; UCI; UCSD; UCB; Stanford; UW	*Various Schools*
2025 Fall	**Invited Talk**, Training Future Proof Developers	*HKU; HKUST; ECNU*
2025 Fall	**Invited Keynote**, AI x Future of Work Series: Workforce Preparation on AI Literacy	*Plug & Play Tech Center*
2025 Fall	**Invited Talk**, Training Requirement-Driven LLM Use for Prompt Programming	*Mila HCAI seminar*
2025	**Invited Talk**, Training the Use of LLMs on Programmatic Data Science for Non-Experts	*Bloomberg; Pitt*
2025 Summer	**Invited Talk**, IJCAI 2025 Sister Conferences' Best Paper Presentation	*Montreal, Canada*
2025 Summer	**Invited Talk**, CSCL/ISLS 2025 IAALDE-Network Best Paper Presentation	*Helsinki, Finland*
2024 Fall	**Invited Talk**, Scaling Student Support: Online Question-Answering with LLM Tutors	*Google DeepMind*
2024 Fall	**Invited Talk**, Training for the Future: Preparing Students for Human-LLM Partnerships	*UMich; CMU; Pitt*

Teaching & Mentoring Experience

2023-present	**Mentor**, Independent Studies or Projects on LLM-assisted Learning Advised students from LS, HCI, CS, and NLP on various projects [P10, P13, P12, W4], now UMich PhD, CMU PhD, etc.	*8 Undergrad/grad; 1 HS*
2024 Fall	**Mentor**, CMU Paths to AI Research Program *for traditionally underrepresented*	*2 Undergraduates*
2022, 2024	**Mentor**, CMU SCS Graduate Application Support Program *for underrepresented*	*3 Master's students*
2023, 2024	**Mentor**, CMU LearnLab Summer School Computing Education Research (e.g., [W5])	*7 Ph.D. and 1 Master's*

2026 Fall	**Guest Lecturer**, CMU 05-315/615 Persuasive Design	*60+ Undergrad/grad*
2026 Spring	**Guest Lecturer**, CMU 05-840 Tools for Online Learning	*25+ Undergrad/grad*
2026 Spring	**Guest Lecturer**, CMU 05-499/899 AI Augmented Designer	*60+ Undergrad/grad*
2025 Fall	**Guest Lecturer**, University of Michigan EECS-498 AI in Education	*15+ Undergrad/grad*
2025 Summer	**Guest Lecturer**, Northeastern INFO-7375 Prompt Engineering for Generative AI	*40+ Undergrad/grad*
2025 Spring	**Guest Lecturer**, CMU 05-898 HCI & Data Science for Product Managers	*40+ Undergrad/grad*
2024 Summer	**Guest Lecturer**, CMU 82-183 AI for Humanities: The Multi-dimensions of World	*20+ Undergraduate*
2024 Fall	**Teaching Assistant**, CMU 05-430/630 Programming Usable Interfaces	*70+ Undergrad/grad*
2023 Fall	**Teaching Assistant**, CMU 05-391/891 Design Human-Centered Software	*60+ Undergrad/grad*
2021 Fall	**Teaching Assistant**, CMU 15-110 Principles of Computing	*300+ Undergraduates*
2021 Summer	**Teaching Assistant**, CMU CS Scholar Pre-College Program *for underrepresented*	*30+ Grades 9-12*
2021 Spring	**Teaching Assistant**, CMU 15-150 Functional Programming (Reflection Blog)	*300+ Undergraduates*

Service & Leadership Experience

2024 - present	**Program Committee**, Various conferences NeurIPS GAIED (2023), EDM (2024-26), SIGCSE (2025), ICLR BiAlign (2025), AIED (2026), ALIT4ALL (2026), NeurIPS BiAlign (2026)	
2022 - present	**Reviewer**, Various conferences and journals ACM CHI (2023-26), UIST (2024-26), Computers & Education (2024) 🏆 *Special Recognitions for Outstanding Reviews* (ACM CHI)	
2019 - present	**Member**, CMU University Disciplinary Committee & Academic Review Board	*Pittsburgh, PA*
2026 - 2027	**Publicity Chair**, Human–AI Interaction Conference (HAIC)	*Washington, DC*
2026 Summer	**Student Volunteer**, CHI (2026)	*Barcelona, Spain*
2026 Summer	**Organizer**, AIED Workshop: 2nd Workshop on AI Literacy Education for All [W3]	*Seoul, South Korea*
2026 Spring	**Panelist**, CHI Workshop: Human-Human Collaboration to Human-Agent Collaboration	*Barcelona, Spain*
2025 Fall	**Invited Expert**, MIT Media Lab Benchmarks for Human Flourishing with AI Workshop	*Boston, MA*
2025 Spring	**Co-leader**, NSF I-Corps Team for LLM-powered Teaching Assistant Project	*Pittsburgh, PA*
2025 Spring	**Member**, HCII NSF Research Experiences for Undergrad (REU) Admissions Committee	*Pittsburgh, PA*
2025 Spring	**Member**, CMU HCII Ph.D. Admissions Committee	*Pittsburgh, PA*
2025 Spring	**Panelist**, K-12 AI Teacher Professional Development at Eberly Center	*Pittsburgh, PA*
2024 Fall	**Panelist**, CMU School of Computer Science Freshman Immigration Course	*Pittsburgh, PA*
2023 Spring	**Director**, CMU School of Computer Science Graduate Student Musical Organize and direct a musical revue with 30+ performers and 200+ audience	*Pittsburgh, PA*
2022 Fall	**Co-leader**, CMU Women@SCS TechNights (data bias session) Help organize a session on training data and a data bias card game for 20+ middle-school Black girls	*Pittsburgh, PA*
2021 Fall	**Member**, CMU Teaching Innovation Award Selection Committee	*Pittsburgh, PA*
2018 - 2022	**President (21-22), VP of Communications (20), Project Advisor (18-19)**, Project Ignite Recruit college students to provide free interdisciplinary project-based learning opportunities to local high-school students, increase diversity in high-school student applications, and facilitate about 100 workshop sessions per year	*Pittsburgh, PA*
2018 - present	**Docent**, Carnegie Museum of Art (CMOA), leading public tours and school tours	*Pittsburgh, PA*